

No 4

16.10.2024

3D

$$V = \begin{vmatrix} x_1 - x_c & y_1 - y_c & z_1 - z_c \\ x_2 - x_c & y_2 - y_c & z_2 - z_c \\ \vdots & \vdots & \vdots \\ x_m - x_c & y_m - y_c & z_m - z_c \end{vmatrix}$$

$$x_c = \frac{1}{m} \sum_{i=1}^m x_i$$

$$y_c = \frac{1}{m} \sum_{i=1}^m y_i$$

$$z_c = \frac{1}{m} \sum_{i=1}^m z_i$$

2D

$$V_1 = \begin{vmatrix} f_1 - f_c \\ \vdots \\ f_m - f_c \end{vmatrix}$$

$$f_c = \frac{1}{m} \sum_{i=1}^m f_i$$

$$\frac{\partial f}{\partial x} = \frac{(V^T V_1)}{(V^T V)}$$

2D:

$$V = \begin{vmatrix} x_1 - x_c & y_1 - y_c \\ x_2 - x_c & y_2 - y_c \\ \vdots & \vdots \\ x_m - x_c & y_m - y_c \end{vmatrix}$$

$$x_c = \frac{1}{m} \sum_{i=1}^m x_i$$

$$y_c = \frac{1}{m} \sum_{i=1}^m y_i$$

$$V_1 = \begin{vmatrix} f_1 - f_c \\ \vdots \\ f_m - f_c \end{vmatrix}$$

$$V_2 = \begin{vmatrix} x_1 - x_c & f_1 - f_c \\ \vdots & \vdots \\ x_m - x_c & f_m - f_c \end{vmatrix}$$

$$f_c = \frac{1}{m} \sum_{i=1}^m f_i$$

3D:

$$V = \begin{vmatrix} x_1 - x_c & y_1 - y_c & z_1 - z_c \\ x_2 - x_c & y_2 - y_c & z_2 - z_c \\ \vdots & \vdots & \vdots \\ x_m - x_c & y_m - y_c & z_m - z_c \end{vmatrix}$$

$$x_c = \frac{1}{m} \sum_{i=1}^m x_i$$

$$z_c = \frac{1}{m} \sum_{i=1}^m z_i$$

$$y_c = \frac{1}{m} \sum_{i=1}^m y_i$$

$$V_1 = \begin{vmatrix} f_1 - f_c & y_1 - y_c & z_1 - z_c \\ f_2 - f_c & y_2 - y_c & z_2 - z_c \\ \vdots & \vdots & \vdots \\ f_m - f_c & y_m - y_c & z_m - z_c \end{vmatrix}$$

$$V_2 = \begin{vmatrix} x_1 - x_c & f_1 - f_c & z_1 - z_c \\ x_2 - x_c & f_2 - f_c & z_2 - z_c \\ \vdots & \vdots & \vdots \\ x_m - x_c & f_m - f_c & z_m - z_c \end{vmatrix}, \dots$$

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$$X_c = \frac{1}{4} \sum_{i=1}^4 x_i$$

$$= \frac{1}{4} (2+6+4+1) = \frac{13}{4}$$

$$y_c = \frac{1}{4} \sum_{i=1}^4 y_i = \frac{1}{4} (1+2+4+3) = \frac{10}{4} = \frac{5}{2}$$

$$V = \begin{bmatrix} x_1 - x_c & y_1 - y_c \\ x_2 - x_c & y_2 - y_c \\ x_3 - x_c & y_3 - y_c \\ x_4 - x_c & y_4 - y_c \end{bmatrix} = \begin{bmatrix} 2 - \frac{13}{4} & 1 - \frac{5}{2} \\ 6 - \frac{13}{4} & 2 - \frac{5}{2} \\ 4 - \frac{13}{4} & 4 - \frac{5}{2} \\ 1 - \frac{13}{4} & 3 - \frac{5}{2} \end{bmatrix} = \begin{bmatrix} -\frac{5}{4} & -\frac{3}{2} \\ \frac{11}{4} & -\frac{1}{2} \\ \frac{3}{4} & \frac{3}{2} \\ -\frac{9}{4} & \frac{1}{2} \end{bmatrix}_{4 \times 2}$$

$$f_c = \frac{1}{4} \sum_{i=1}^4 f_i = \frac{1}{4} (5 + (-1) + 2 + 4) = \frac{10}{4} = \frac{5}{2}$$

$$\begin{bmatrix} f_1 - f_c \\ f_2 - f_c \\ f_3 - f_c \\ f_4 - f_c \end{bmatrix} = \begin{bmatrix} 5 - \frac{5}{2} \\ -1 - \frac{5}{2} \\ 2 - \frac{5}{2} \\ 4 - \frac{5}{2} \end{bmatrix} = \begin{bmatrix} \frac{5}{2} \\ -\frac{7}{2} \\ -\frac{1}{2} \\ \frac{3}{2} \end{bmatrix}_{4 \times 1} \rightarrow V_1 = \begin{bmatrix} \frac{5}{2} & -\frac{3}{2} \\ -\frac{7}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \\ \frac{3}{2} & \frac{1}{2} \end{bmatrix}_{4 \times 2}, V_2 = \begin{bmatrix} -\frac{5}{4} & \frac{5}{2} \\ \frac{11}{4} & -\frac{7}{2} \\ \frac{3}{4} & -\frac{1}{2} \\ -\frac{9}{4} & \frac{3}{2} \end{bmatrix}_{4 \times 2}$$

$$V^T V_1 = \begin{bmatrix} -\frac{33}{2} & \frac{1}{2} \\ -2 & 5 \end{bmatrix}, V^T V_2 = \begin{bmatrix} \frac{59}{4} & -\frac{33}{2} \\ \frac{1}{2} & -2 \end{bmatrix}, V^T V = \begin{bmatrix} \frac{59}{4} & \frac{1}{2} \\ \frac{1}{2} & 5 \end{bmatrix}$$

$$\det(V^T V_1) = -81.5$$

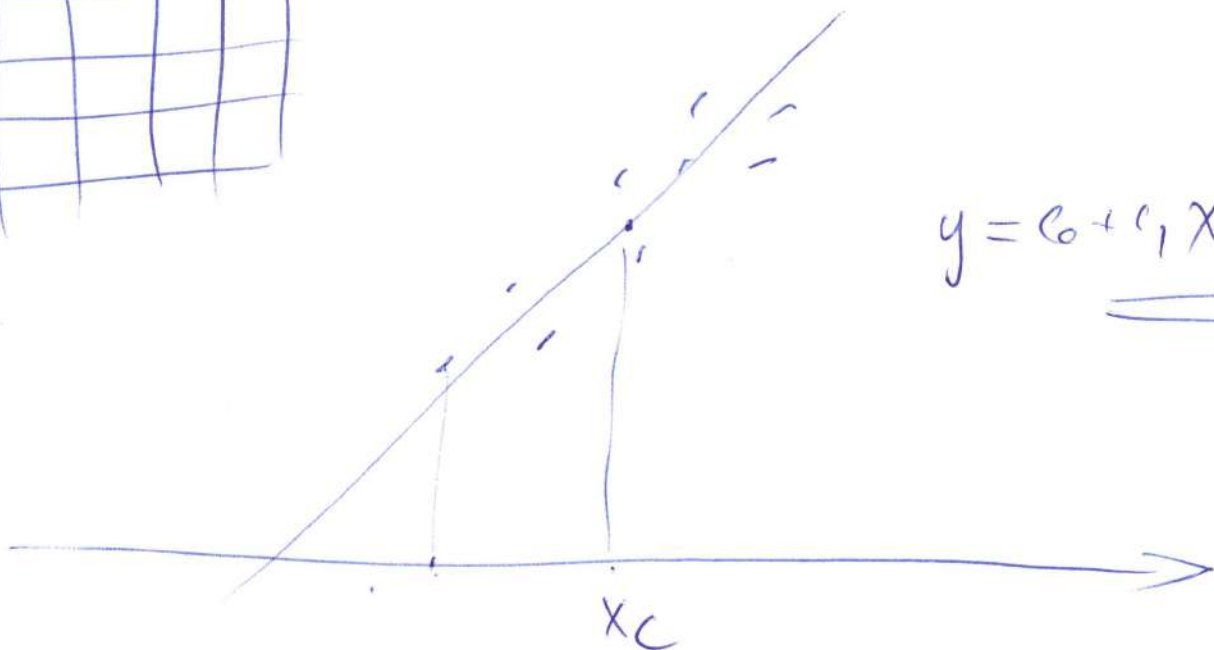
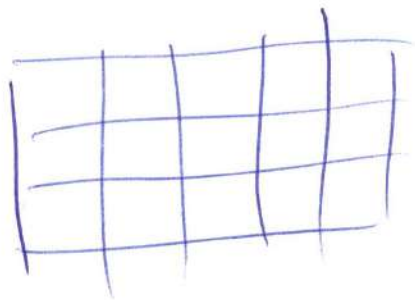
$$\det(V^T V_2) = -21.25$$

$$\det(V^T V) = 73.5$$

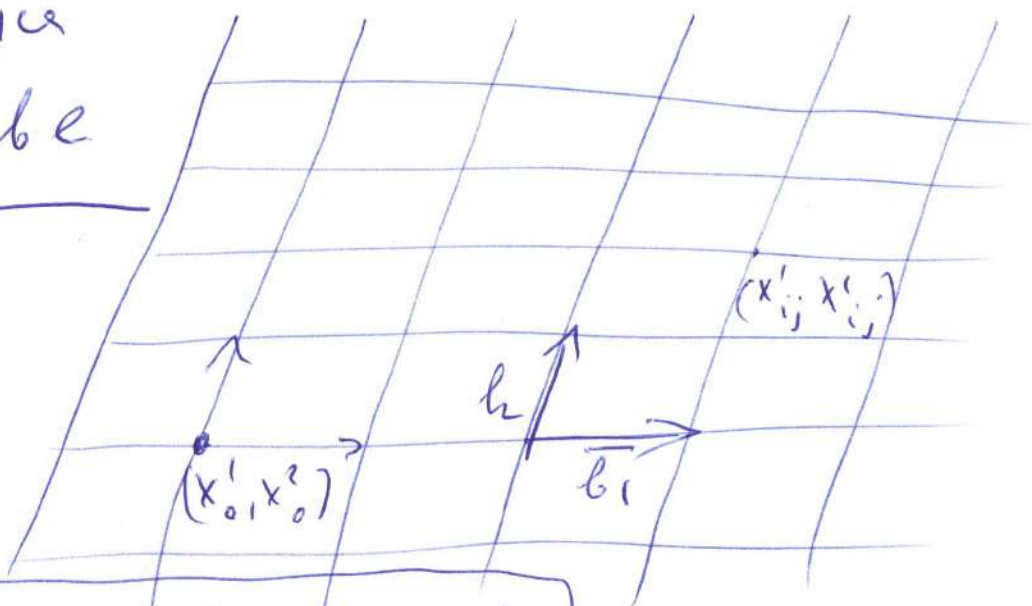
$$\frac{\partial f}{\partial x} \approx \frac{\det(V^T V_1)}{\det(V^T V)} = \frac{-81.5}{73.5} \approx -1.109$$

$$\frac{\partial f}{\partial y} \approx \frac{\det(V^T V_2)}{\det(V^T V)} = \frac{-21.25}{73.5} \approx -0.289$$

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параметры
Таблицы



$$\begin{pmatrix} x'_{ij} \\ x''_{ij} \end{pmatrix} = \begin{pmatrix} x'_0 \\ x''_0 \end{pmatrix} + B_h \begin{pmatrix} i \\ j \end{pmatrix}$$

$$B_h = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \rightarrow \square$$

$$\begin{pmatrix} b_{11} \\ b_{21} \end{pmatrix} = \overline{b_1}$$

$$\begin{pmatrix} b_{12} \\ b_{22} \end{pmatrix} = \overline{b_2}$$

$$B_h = \begin{pmatrix} h_1 & 0 \\ 0 & h_2 \end{pmatrix} \rightarrow \square$$

(в зависимости от) оптимизации

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Сетка

$$\begin{pmatrix} x_{i_1 \dots i_n}^1 \\ \vdots \\ x_{i_1 \dots i_n}^n \end{pmatrix} = \begin{pmatrix} x_o^1 \\ \vdots \\ x_o^n \end{pmatrix} + B_n \begin{pmatrix} i_1 \\ i_2 \\ \vdots \\ i_n \end{pmatrix}$$

$$(i) = (i_1, \dots, i_n) \Rightarrow \bar{x}_{(i)} = \bar{x}_o + B_n \cdot (i)$$

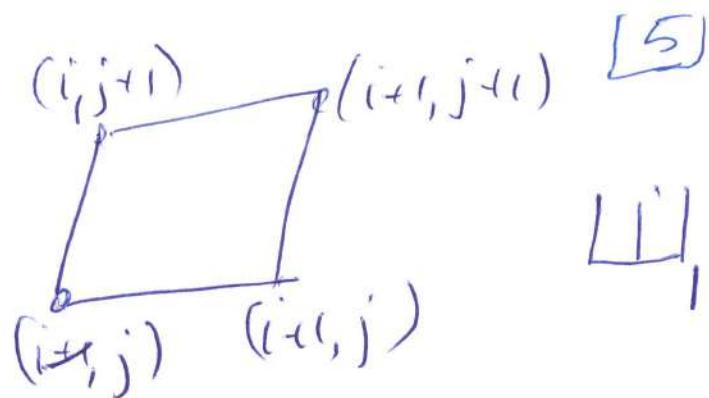
мультимножество

Пусть в узлах - значения функции f
 $\{f(i)\}$ - сеточная функция
сеточная функция

Норм. функции: $f \rightarrow \frac{\partial f}{\partial x_i}$ $\{f(i)\} \rightarrow \bar{U}_n$
 сет. функции $f \rightarrow d_i f$

Оператор $d: V_n \rightarrow \bar{U}_n$

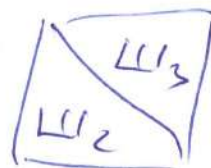
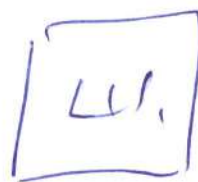
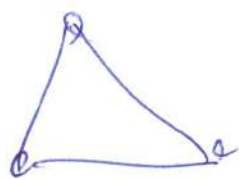
$$[d f]_{(i)} = \sum_{(j) \in \mathcal{U}_1} \beta_{(i,j)} f_{(i)+(j)}$$



$$\mathbb{L}_1 = \{(0,0), (1,0), (1,1), (0,1)\}$$

$$(d + t)_{(i,j)} = \beta_{00} t_{ij} + \beta_{10} t_{i+1,j} + \beta_{11} t_{i+1,j+1} + \beta_{01} t_{i,j+1}$$

$$\mathbb{L}_2 = \{(0,0), (1,0), (0,1)\}$$



$$\mathbb{L}_3 = \{(0,0), (1,1), (0,1)\}$$

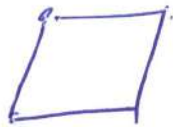
Ansatz equare $(T_{ij} t)_{(i)} = t_{(i)+1,j}$

$$T_{(2,3)} t_{(i)} = t_{(i+2,j+3)} \Rightarrow$$

$d = \beta_{00} T_{00} + \beta_{10} T_{10} + \beta_{11} T_{11} + \beta_{01} T_{01}$ um
admissen system

$$d = \sum_{(j) \in \mathbb{L}_1} \beta_{(j)} T_{(j)}$$

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$$\left. \begin{aligned} d_1^+ f &\approx \frac{\partial f}{\partial x} \\ d_2^+ f &\approx \frac{\partial^2 f}{\partial x^2} \end{aligned} \right\} \begin{aligned} &\text{p-loc} \\ &\text{ест. оператор} \\ &\text{приближение} \\ &= \beta_i; \end{aligned}$$

$$d_0^+ f \approx f$$

Сопреженный оператор

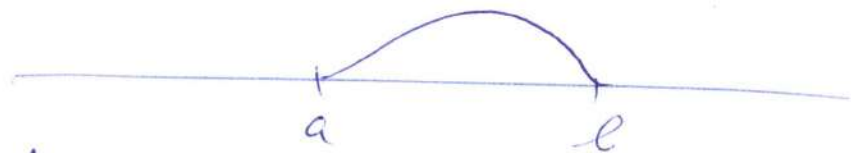
$$\int_a^b f(x) \frac{\partial g}{\partial x} dx = f(b)g(b) - f(a)g(a) - \int_a^b \frac{\partial f}{\partial x} g(x) dx$$

$f(x), g(x)$ - функции

$f(x), g(x) = 0$ на концах (a, b)

$f(x), g(x) \in C^1(-\infty, +\infty)$

$f(x) \neq 0$
 $g(x) = 0$ if $f, g \in [a, b]$
финитные функции



$$\Rightarrow \int_a^b f(x) \frac{\partial g}{\partial x} dx = - \int_a^b \frac{\partial f}{\partial x} g(x) dx$$

X, y - векторы

(x, y) - скалярное произведение

A - оператор, A^* - сопряженный оператор,

$$\text{если } (Ax, y) = (x, A^*y)$$

скалярное произведение

элемент

[7]

f, g - continuous functions

$$(f, g) = \int_a^b f(x) g(x) dx$$

$$(f_1 + f_2, g) = (f_1, g) + (f_2, g)$$

$$\frac{d}{dx} : \int_a^b f(x) \frac{dg}{dx} dx = - \int_a^b \frac{df}{dx} g(x) dx \quad (*)$$

$$\begin{aligned} & \uparrow \\ & (g, f) = - (f, g) \end{aligned}$$

$$A = \frac{d}{dx} \Rightarrow A^* = - \frac{d}{dx} \quad \left| \left(\frac{d}{dx} \right)^* = - \frac{d}{dx} \right.$$

$$\begin{array}{c} | \quad | \quad | \quad | \quad | \quad | \quad | \\ a = x_0 \qquad \qquad \qquad x_{n-1} \quad b = x_n \end{array}$$

$$\int_a^b \approx \sum_{i=0}^{n-1} h \left(f_i \cdot \frac{g_{i+1} - g_i}{h} \right) = \sum_{i=0}^{n-1} f_i (g_{i+1} - g_i) = \begin{cases} x_i = a + i h \\ h = \frac{b-a}{n} \end{cases}$$

$$\begin{aligned} &= \underbrace{(-f_0 g_0) + f_0 g_1 - f_1 g_1 + f_1 g_2 - f_2 g_2 + f_2 g_3 - \dots}_{\dots - f_{n-2} g_{n-2} + f_{n-2} g_{n-1} - f_{n-1} g_{n-1} + f_{n-1} g_n} = \\ &= f_{n-1} g_n - f_0 g_0 + g_1 (f_0 - f_1) + g_2 (f_1 - f_2) + \dots + g_{n-1} (f_{n-2} - f_{n-1}) \end{aligned}$$

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$$= t_{n-1}g_n - t_0g_0 - g_1(t_1 - t_0) - g_2(t_2 - t_1) - \dots - g_{n-1}(t_n - t_{n-1})$$

$$= t_{n-1}g_n - t_0g_0 - \sum_{i=1}^{n-1} g_i(t_i - t_{i-1}) =$$

$$= t_{n-1}g_n - t_0g_0 - \sum_{i=1}^{n-1} h g_i \left(\frac{t_i - t_{i-1}}{h} \right)$$

$$\sum_{i=0}^{n-1} h \left(t_i, \frac{g_{i+1} - g_i}{h} \right) = t_{n-1}g_n - t_0g_0 - \sum_{i=1}^{n-1} h g_i \left(\frac{t_i - t_{i-1}}{h} \right)$$

$$\int_a^b f(x) \frac{\partial g}{\partial x} dx = f(b)g(b) - f(a)g(a) - \int_a^b \frac{\partial f}{\partial x} g(x) dx$$

f, g - functions $\Rightarrow f_0 = f(a) = 0, g_0 = 0$
 $f_n = 0, g_n = 0$

$$\Rightarrow \left[\sum_{i=0}^{n-1} h t_i \left(\frac{g_{i+1} - g_i}{h} \right) = - \sum_{i=0}^{n-1} h \left(\frac{t_i - t_{i-1}}{h} \right) g_i \right] \leftarrow \text{аналог } (x)$$

Правда $(d_1^+ t)_i = \frac{1}{h}(t_{i+1} - t_i)$ - правый элемент

$(d_1^- t)_i = \frac{1}{h}(t_i - t_{i-1})$ - левый элемент

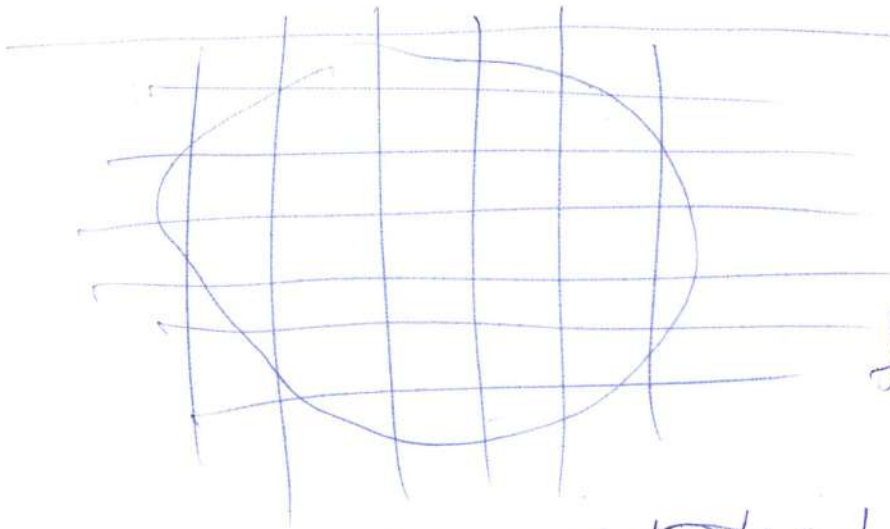
$$(t, g) = h \sum_{i=1}^n t_i g_i \Rightarrow (t, d_1^+ g) = -(d_1^- t, g)$$

$$\Rightarrow (d_1^+)^* = -d_1^-$$

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Смешанный оператор сдвига

Ther 6 $T_{(j)}$ - inverse of cyl_{inv} , $\Rightarrow [T_{(j)}]^* = T_{-(j)}$



2-5-18 RZ

$$\Omega_n = \{ (i) \mid \bar{X}_{(i)} \in \Omega \}$$

$f(\bar{x}), g(\bar{x})$ - функции
в аргументе $\bar{x}$ и $\bar{U}$

$$(f, g)_{\overline{V}} = \int_{-2}^2 f(\overline{x}) g(\overline{x}) d\overline{x} d\overline{y} \quad \text{for } f, g \in \overline{V}$$

Get er in

Ag, Ag⁺ - Ag^+ - Ag^+

в функ. н-е U_h
функционал. прост U_h

$$(f, s)|_{U_n} = \sum_{(i,j) \in \mathbb{Z}_n} \Delta s'_{ij} g_{ij}$$

28985 Snow: SD

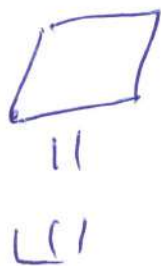
~~$$(T_{(1)} + 1, g) \cdot U_n = \Sigma$$~~

$$(T_{(i, e)} + g) / U_h = \sum_{(i, j) \in R_h} \Delta S_{i, i+1, j+1} g_{ij} = \sum_{(m, n) \in (?) } \Delta S_{m, m+1, n+1} g_{m-1, n-1}$$

$$= (f, T_{(k, e)} s) \quad \boxed{\square} \quad \mathbb{R}_h$$

$$\Rightarrow T_{(11, e)}^* = T_{(11, e)}$$

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$$\begin{aligned} d_0^+ &\sim \mathbf{I} & d_0^+ &= \sum_{(k, \ell) \in LI} \beta_{k\ell}^0 T_{k\ell} \\ d_1^+ &\sim \frac{d}{dx} & d_1^+ &= \sum_{(k, \ell) \in LI} \beta_{k\ell}^1 T_{k\ell} \\ d_2^+ &= \frac{d}{dy} & d_2^+ &= \sum_{(k, \ell) \in LI} \beta_{k\ell}^2 T_{k\ell} \end{aligned}$$

$$(A+B)^* = A^* + B^*$$

$$(cA)^* = c \cdot A^*$$

$$(d_0^+)^* = \sum_{(k, \ell) \in LI} \beta_{k\ell}^0 T_{(-k, -\ell)}$$

$$(d_1^+)^* = \sum \beta_{k\ell}^1 T_{(-k, -\ell)} \approx -\frac{d}{dx}$$

$$(d_2^+)^* = \sum \beta_{k\ell}^2 T_{(-k, -\ell)} \approx -\frac{d}{dy}$$

$I^* = I$ - тождественная операция

$$(IA, B) = (A, B) = (A, IB)$$

$$\left(\frac{d}{dx} A, B\right) = -\frac{d}{dx} (A, B)$$

$$d_0^- = (d_0^+)^*$$

$$d_1^- = -(d_1^+)^*$$

$$d_2^- = -(d_2^+)^*$$

$$(d_m^+ t)_{ij} = \sum_{(k, \ell) \in LI} \beta_{k\ell}^m t_{i-k, j+\ell}$$

$$(d_m^- t)_{ij} = \chi_m \sum_{(k, \ell) \in LI} \beta_{k\ell}^m t_{i-k, j-\ell}$$

$$\chi_m = (-1)^k$$

k - номер нулевого члена

$$\chi_m = \begin{cases} 1 & m=0 \\ -1 & m=1 \\ -1 & m=2 \end{cases}$$

Сторонне переписываем интегральное равенство

$$\int_{\Omega} f(x,y) \frac{\partial g}{\partial x} dx dy = - \int_{\Omega} \frac{\partial f}{\partial x} g(x,y) dx dy \quad \left(\begin{array}{l} \text{формула} \\ \text{Грина} \end{array} \right)$$

Сторонний аналог:

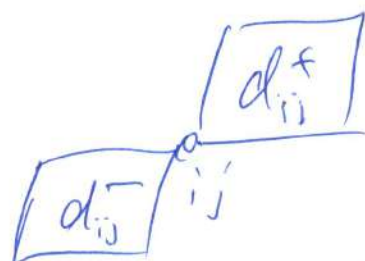
$$\sum_{(i,j) \in \Omega_h} f_{ij} (d_i^+ g)_{ij} \Delta S = - \sum_{(i,j) \in \Omega_h} (d_i^- f)_{ij} g_{ij} \Delta S$$

$$\sum_{(i,j) \in \Omega_h} f_{ij} (d_i^+ g)_{ij} \Delta S = - \sum_{(i,j) \in \Omega_h} (d_i^- f)_{ij} g_{ij} \Delta S$$

$$\sum_{(i,j) \in \Omega_h} f_{ij} (d_i^+ g)_{ij} \Delta S = \sum_{(i,j) \in \Omega_h} (d_i^- f)_{ij} g_{ij} \Delta S$$



$$\int f \cdot g d\Omega = \int f \cdot g d\Omega$$



Формула Эйлера-Биринга

Формула Грина в двумерном
пространстве $\mathbb{R}^2$

$$W = \int_{\Omega} F(u, p_1, p_2) d\Omega$$

$$p_1 = \frac{\partial u}{\partial x^1}, p_2 = \frac{\partial u}{\partial x^2}$$

Вариационная задача

Вариационная задача. Найти u , при котором W

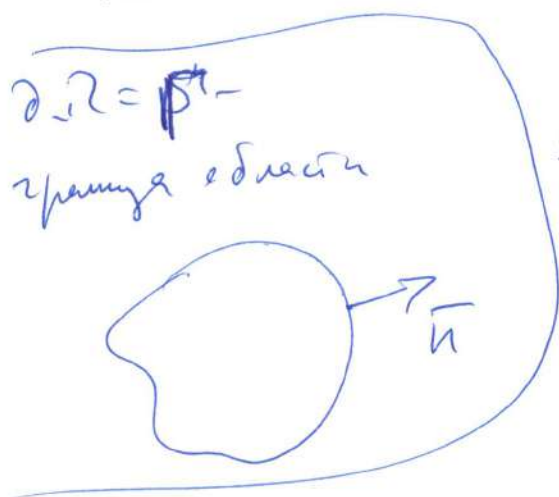
принимает минимальное значение.

$$\delta W = \int_{\Omega} \frac{\partial F}{\partial u} \delta u + \frac{\partial F}{\partial p_1} \delta p_1 + \frac{\partial F}{\partial p_2} \delta p_2 d\Omega = 0$$

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$$\delta W = \int_{\Omega} \frac{\partial F}{\partial u} \delta u + \frac{\partial F}{\partial p_1} \frac{\partial (\delta u)}{\partial x^1} + \frac{\partial F}{\partial p_2} \frac{\partial (\delta u)}{\partial x^2} d\Omega =$$

$$= \int_{\partial\Omega} \frac{\partial F}{\partial p_1} n_1 \delta u + \frac{\partial F}{\partial p_2} n_2 \delta u + \int_{\Omega} \frac{\partial F}{\partial u} \delta u - \frac{\partial}{\partial x^1} \frac{\partial F}{\partial p_1} \delta u - \frac{\partial}{\partial x^2} \frac{\partial F}{\partial p_2} \delta u d\Omega$$



$$= \int_{\partial\Omega} \left(\frac{\partial F}{\partial p_1} n_1 + \frac{\partial F}{\partial p_2} n_2 \right) \delta u d\Gamma +$$

$$+ \int_{\Omega} \left(\frac{\partial F}{\partial u} - \frac{\partial}{\partial x^1} \frac{\partial F}{\partial p_1} - \frac{\partial}{\partial x^2} \frac{\partial F}{\partial p_2} \right) \delta u d\Omega = 0$$

$$\frac{\partial F}{\partial p_1} n_1 + \frac{\partial F}{\partial p_2} n_2 = 0 \text{ на границе}$$

на границе

$$\frac{\partial F}{\partial u} - \frac{\partial}{\partial x^1} \frac{\partial F}{\partial p_1} - \frac{\partial}{\partial x^2} \frac{\partial F}{\partial p_2} = 0 \text{ внутри области}$$

внутри области

Уравнение Эйлера вариационной задачи

$\delta = \cos + i \sin$